AE 452 Soil and Water Engineering II (2,3,3)

Lecturer: Dr. Wilson A. Agyare

Rainfall Runoff and Stream flow

Rainfall Runoff

Rainfall Runoff

- Runoff results from rainfall occurrence in a hydrologic catchment.
- Studies of runoff prediction (to estimate volume and peak runoff flows) usually requires past records.
- Factors influencing runoff generation
  - Soil
    - infiltration capacity
    - level of capillary storage (or soil moisture storage limit)
  - Land surface condition
    - Vegetation
    - Topography
  - Catchment area

Rainfall Runoff

Problems of estimating runoff from rainfall are due to the :

- absence of stream flow data (gauges)
- absence of recording rain gauges - to provide rainfall intensities data (i.e. the use of non-recording gauges that provides only rainfall amount).

Rainfall Runoff

- **Time of Concentration (Tc):** It is the longest time it takes for a part of the catchment to contribute water to the outlet. It is the time it takes for all the parts of the watershed to be contributing water to the outlet.
- Some empirical methods are available for predicting runoff in a catchment without the stream flow and recording gauge information.

Runoff by Rational method

$$Q_p = (CIA)/360$$

- Where Q_p is the peak flow (m^3 /s);
- C is dimensionless runoff coefficient dependent on the catchment characteristics;
- I is the mean rainfall intensity (mm/h) for a given return period or recurrence interval (i.e. the average number of years within which a given rainfall event will be expected to occur at least once)
  - Assumption –
    - rate of rainfall is constant during T_c (time of concentration)
    - All the measured rainfall in the area contributes to the flow.
- A is the area of catchment (ha)

Using the Rational method

- Obtain area of catchment by surveying, satellite imagery, from maps or aerial photographs.
- Estimate intensity using rainfall intensity curves.
- The runoff coefficient C is the fraction of rainfall which becomes runoff. On a corrugated iron roof, almost all the rain would runoff so C = 1, while in a well drained soil, nine-tenths of the rain may soak in and so C = 0.10.
  - These values will have to be obtain from experiment or literature.
  - For catchments with different kinds of characteristics, the different values should be combined in proportion to their area

Using the Rational method

Mean catchment characteristic

Example: For a given area A, made up of area A1, A2, A3…An, with runoff coefficients C1, C2, C3…Cn, then

$$C = \frac{\sum_{n=1}^{n} CnAn}{\sum_{n=1}^{n} An}$$

Using the Rational method

Values of run-off coefficient C

Topography and vegetation	Soil texture		
	Open sandy loam	Clay and silt loam	Tight clay
Woodland			
Flat 0-5 per cent slope	0·10	0·30	0·40
Rolling 5-10 per cent slope	0·25	0·35	0·50
Hilly 10-30 per cent slope	0·30	0·50	0·60
Pasture			
Flat	0·10	0·30	0·40
Rolling	0·16	0·36	0·55
Hilly	0·22	0·42	0·60
Cultivated			
Flat	0·30	0·50	0·60
Rolling	0·40	0·60	0·70
Hilly	0·52	0·72	0·82
Urban areas	30 per cent of area impervious	50 per cent of area impervious	70 per cent of area impervious
Flat	0·40	0·55	0·65
Rolling	0·50	0·65	0·80

From Schwab, Frevert, Edminster, and Barnes, *Soil and water conservation engineering*, Wiley, New York.

Runoff by Cook's method

Factors considered:

- catchment characteristics (cover type, soil type and drainage, and slope)
- area of catchment
- shape of catchment
- return period

Using the Cook's method

1. Obtain the catchment characteristics (CC) base on vegetation, soil type and drainage, and slope from Table
2. Obtain runoff based on CC and Area
3. Modify runoff based on the **return period** and **shape of area**

Using the Cook's method

Example:

A catchment made up of heavy grass (10) on shallow soils with impeded drainage (30) and moderate slope (10). Therefore the catchment characteristics (CC) is then the sum of the three i.e. 50. Then using the catchment area of 100 ha and the CC, the maximum runoff can be read from Table as 10.4 m^3s^{-1}. This gives the runoff for a 10 year return period.

Using the Cook's method

Catchment characteristics (CC)

Cover		Soil type and drainage		Slope	
Heavy grass	10	Deep, well drained soils	10	Very flat to gentle	5
Scrub or medium grass	15	Deep, moderately pervious soil	20	Moderate	10
Cultivated lands	20	Soils of fair permeability and depth	25	Rolling	15
Bare or eroded	25	Shallow soils with impeded drainage	30	Hilly or steep	20
		Medium heavy clays or rocky surfaces	40	Mountainous	25
		Impervious surfaces and waterlogged soils	50		

Select the most appropriate factor from each of these three lists and add them together.

Example

Heavy grass (10) on shallow soils with impeded drainage (30) and moderate slope (10):
CC = 10 + 30 + 10 = 50.

From Dept. of Conservation and Extension, Government of Rhodesia. *Handbook of basic instruction for dam construction.*

Using the Cook's method

Area in ha and Runoff in m^3s^{-1}

Run-off from small catchments

CC \	25	30	35	40	45	50	55	60	65	70	75	80
5	0·2	0·3	0·4	0·5	0·7	0·9	1·1	1·3	1·5	1·7	1·9	2·1
10	0·3	0·5	0·7	0·9	1·1	1·4	1·7	2·0	2·4	2·8	3·2	3·7
15	0·5	0·8	1·1	1·4	1·7	2·0	2·4	2·9	3·4	4·0	4·6	5·2
20	0·6	1·0	1·4	1·8	2·2	2·7	3·2	3·8	4·4	5·1	5·8	6·5
30	0·8	1·3	1·8	2·3	2·9	3·6	4·4	5·3	6·3	7·3	8·4	9·5
40	1·1	1·5	2·1	2·8	3·5	4·5	5·5	6·6	7·8	9·1	10·5	12·3
50	1·2	1·8	2·5	3·5	4·6	5·8	7·1	8·5	10·0	11·6	13·3	15·1
75	1·6	2·4	3·6	4·9	6·3	8·0	9·9	11·9	14·0	16·4	18·9	21·7
100	1·8	3·2	4·7	6·4	8·3	10·4	12·7	15·4	18·2	21·2	24·5	28·0
150	2·1	4·1	6·3	8·8	11·6	14·7	18·2	21·8	25·6	29·9	35·0	40·6
200	2·8	5·5	8·4	11·7	15·3	19·1	23·3	28·0	33·1	38·5	45·0	52·5
250	3·5	6·5	9·7	13·2	17·2	21·7	27·0	32·9	39·6	46·9	55·0	63·7
300	4·2	7·0	10·5	14·7	19·6	25·2	31·5	38·5	46·2	54·6	63·7	73·5
350	4·9	8·4	12·6	17·2	23·2	30·2	37·8	46·3	55·8	62·5	73·5	81·0
400	5·6	10·0	14·4	19·4	25·6	33·6	42·2	51·0	60·0	69·3	79·5	90·0
450	6·3	10·5	15·5	21·5	28·5	36·5	45·5	55·5	65·5	76·0	86·5	97·5
500	7·0	11·0	17·0	23·5	31·0	40·5	51·0	62·0	73·0	84·0	95·0	106·5

From Dept. of Conservation and Extension, Government of Rhodesia, Dept. In-Service Manual.

Using the Cook's method

Return period: for return periods different from 10 years, the conversion factor is:

	Return period and conversion factors				
Return Period (yrs)	2	5	10	25	50
Conversion factor	0.90	0.95	1.00	1.25	1.50

Catchment shape factor: Runoff for a almost square or round catchment is **1**. For other shapes, use the conversion factors:

	Catchment Shape and factor		
Catchment Shape	Square or round	Long & narrow	Broad & short
Factor	1	0.8	1.25

Stream flow

Stream flow

Aim of measuring stream flow: is to determine the discharge

Stage: Height of stream level measured from an arbitrary datum (this can be the mean sea level (msl))

Aim of measuring stage: is mainly to establish a stage-discharge relationship

Depth: Height of stream measured from the bottom of the channel.

Stage measurements

1. **Manual**

a. **Staff Gauges**: This is fixed vertically to the river bank at a stable point where it is not affected by turbulence or wave action.

b. **Section staff gauge:** made of a series of posts each overlapping. The height above a certain datum is measured.

c. Datum level set on a bridge and a string having a weight electrically connected is lowered to the water surface to obtain the depth of water surface.

Stage Measurement

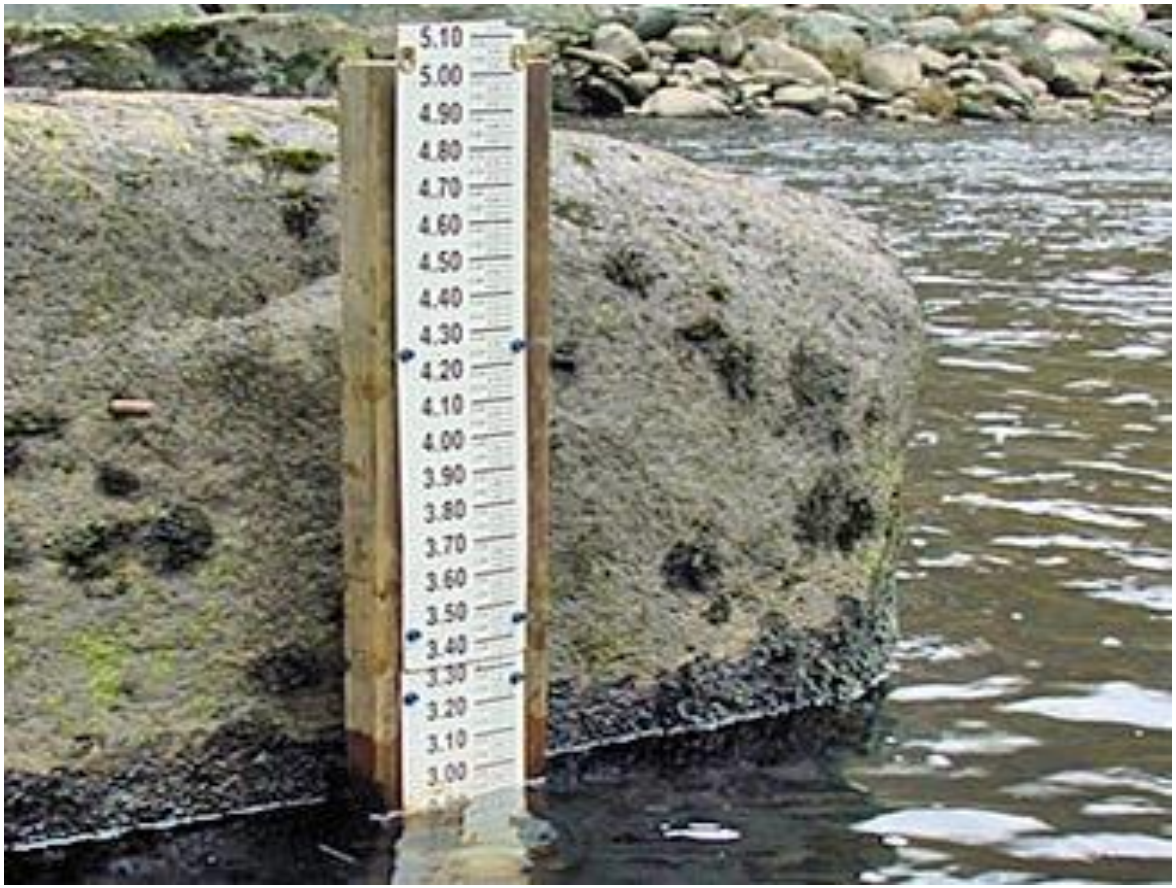


Figure: Sample of staff gauge

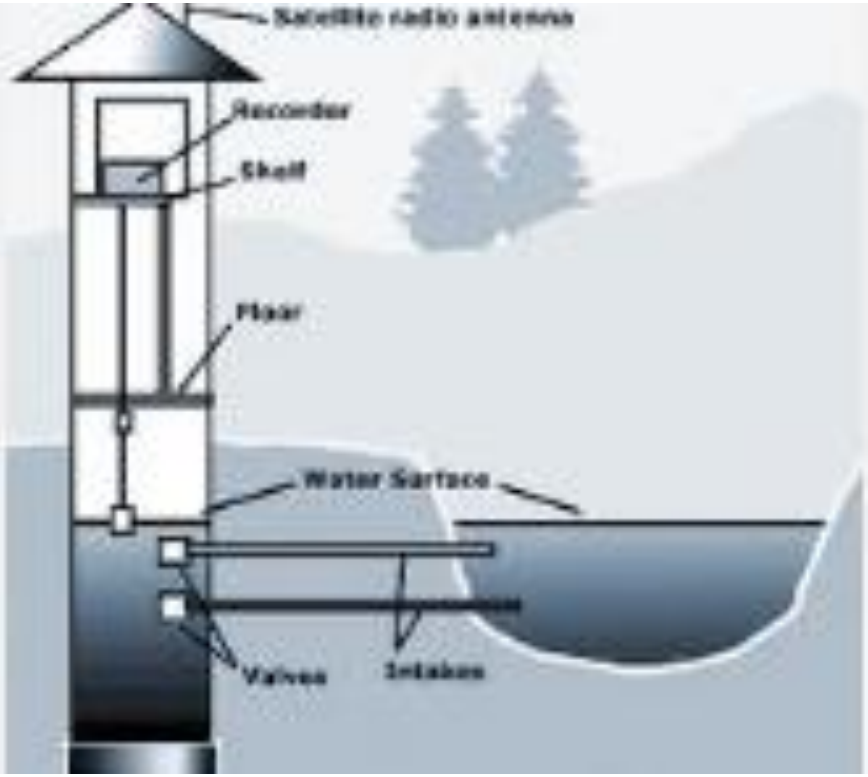
Stage measurements

2. Recording gauges

a. **Float Gauge:** Float movement fluctuates with change in stage and this is recorded by a chart. Both the high and low flows are measured on the chart.

b. **Digital Recorders:** They have clocks and at the set time interval (eg. 30-min) measurements are recorded


USGS


Satellite radio antenna
Recorder
Shelf
Float
Water Surface
Valves
Intakes

Float gauge



Float gauge mechanism

Stage measurements

2. Recording gauges

c. **Crest Gauges:** Measure only peak flows. It is a made of cylindrical tube sealed below with only a few holes to allow the water to enter the tube. A ground cork fixed in the tube floats up and is held by surface tension when stage increases. It stays at maximum stage until the reading is taken and let loose.

d. **Pressure gauge:** The measurement of stage is by pressure transducers, an indirect method converting the hydrostatic pressure of a submerged datum to water level above. A typical e.g. is the diver capable of recording pressure levels over time which is later downloaded. Very useful for gauging stations where it is impractical to build stilling wells for float gauges.

Rules guiding location of gauges

a. Gauges should not be located in rivers with scouring characteristics
b. Gauge locations should not be close to river bends because the water surface is usually inclined at bends and also there is turbulence making the stage measurement inconsistent
c. Locate gauge at where channel section is uniform to obtain good stage measurement (i.e. avoid irregular cross sections).

Velocity Measurements

Current Meter: It consists of a propeller which rotates when water passes through it. It records the propeller rotation per unit time.

- The number of rotations are correlated to velocity of flow using the formula:

  $V = a + bN$

  – where N is the number of rotations of the propeller (revs per second)

  – a and b are coefficients determined by calibration in an experimental flume

Current meter





Velocity Measurements

Depth of Current Meter measurement

1. **Single point method** - average velocity measured at 0.6 of the depth from the surface (i.e. when depth is < 0.76 m)

i.e. V = average velocity is at about 0.6D.

2. **Two-point method** - velocity is taken at 0.2 and 0.8 depths from the surface and the average calculated, when depth is > 0.76 m

$$V = (V_{0.2D}+V_{0.8D})/2$$

Note: The two-point method is more accurate but in a shallow stream cross-section, the velocity at 0.2D may be difficult to measure so a single value at 0.6D

Determination of Discharge

- Divide the cross-section of the stream into about 20 vertical sections such that no section carries more than 10 % of the total flow.
- Take soundings to determine various depths.
- The sections are of known width, so that the discharge can be calculated if the velocities are taken at appropriate depth (0.2D and 0.8D **OR** 0.6D alone).

Determination of Discharge

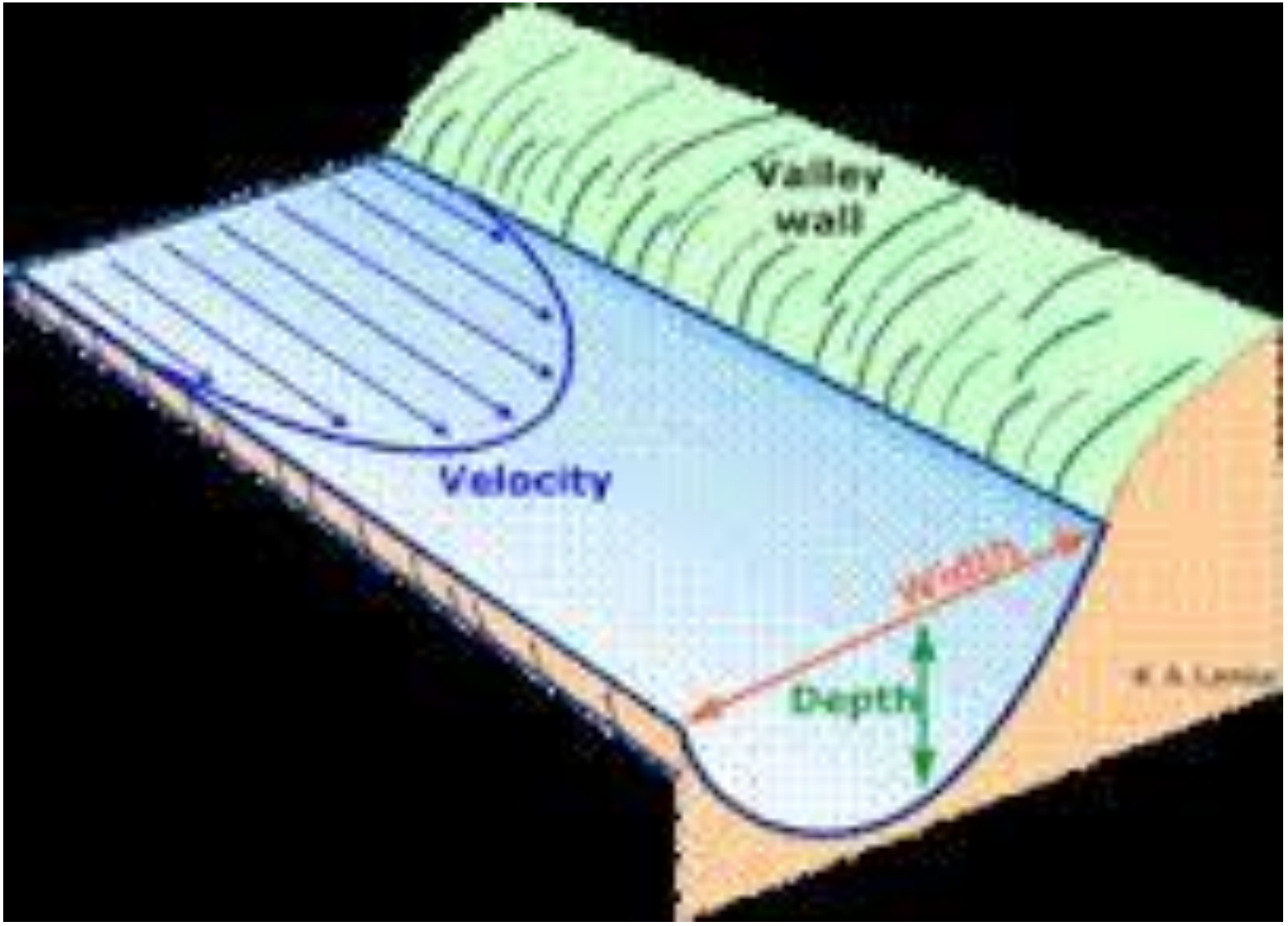
- Discharge in one segment;

  q = Average velocity ($\overline{v}$) × Area of segment (a_i)

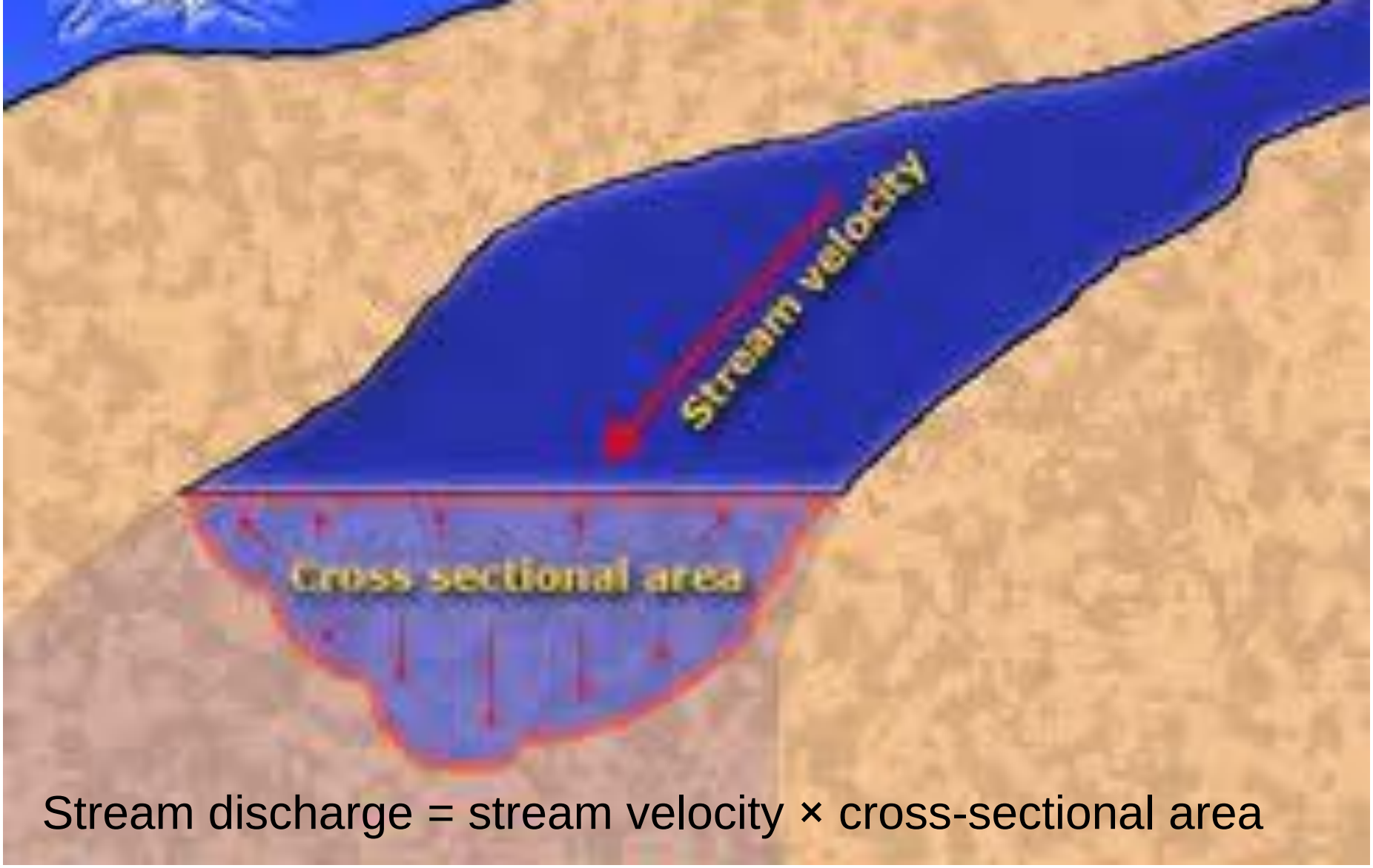
- Area of each segment can be calculated using the trapezoidal formula

- Total discharge,

$$Q = \sum_{i=1}^{n} \overline{v}_i a_i$$



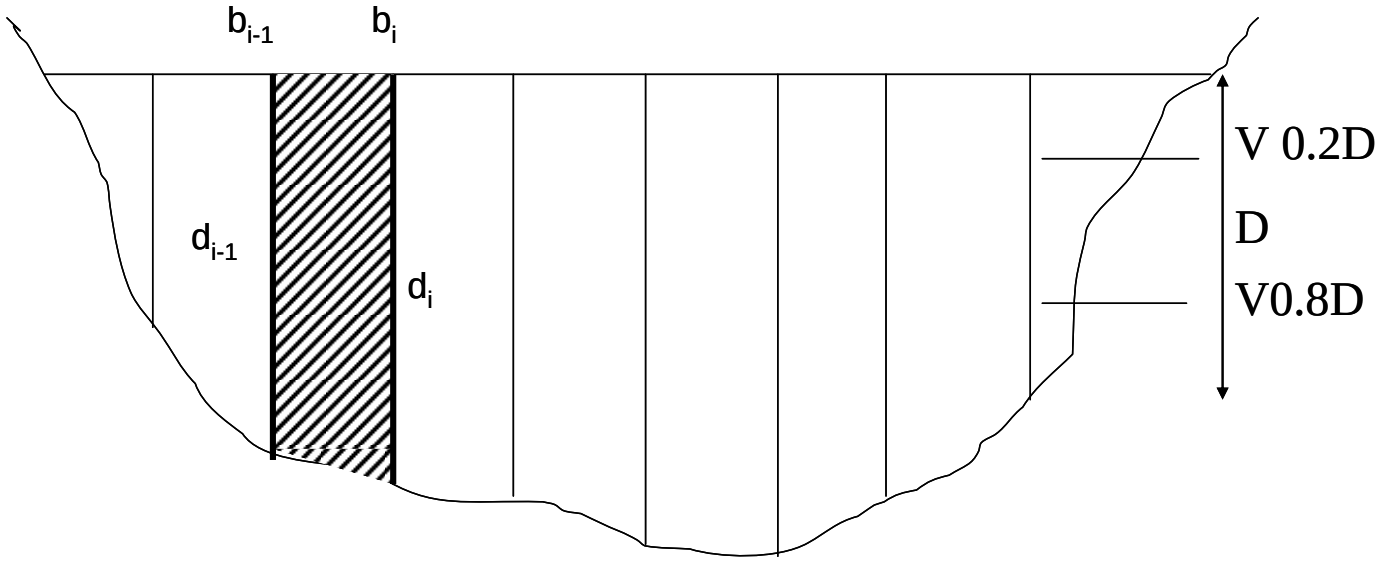
Stream cross-section



Stream discharge = stream velocity × cross-sectional area

Determination of Discharge

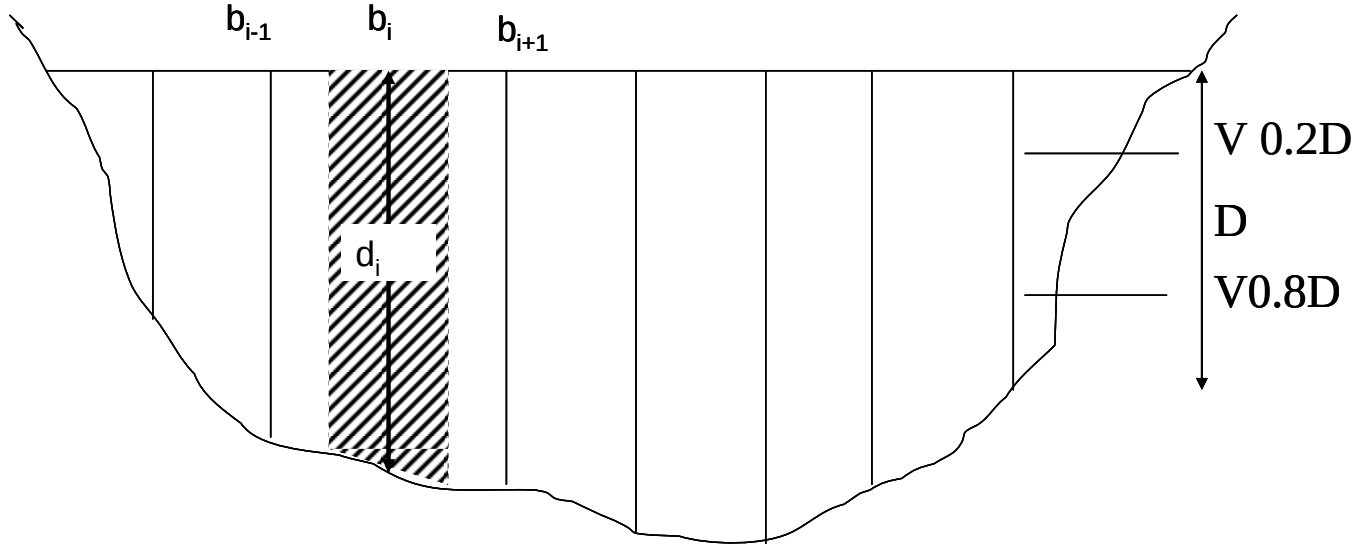
1. Area calculation: Mean-section method



$$A_i = \left(\frac{d_{i-1} + d_i}{2} \right) \left(b_i - b_{i-1} \right)$$

Determination of Discharge

2. Area calculation: Mid-section method



$$A_i = d_i \frac{b_{i+1} - b_{i-1}}{2}$$

Determination of Discharge

Note the following:

- For small streams use pygmy meters
- For streams with steep gradient and high velocity use the dilution method
- For large streams regularly check reading of staff during measurement

Discharge Measurements

Using Floats

- Any floatable substance eg. a tennis ball/cork is placed at a point and the time (t) it takes it to move a known distance (d) is noted
- d/t gives the average surface velocity of the water
- The surface velocity(v_s) is equal to 1.2 (Average velocity, $\bar{v}$) ie. $v_s = 1.2\,\bar{v}$ and $\bar{v} = 0.8\,v_s$
- The cross-sectional area of flow is then multiplied by the average velocity to obtain the discharge

Stage-Discharge Relations

- Simultaneous measurements of stage and discharge provide a calibration graph known as stage-discharge relation or rating curve
- A plot of stage against discharge is made to obtain a rating curve
- The essence of the rating curve is to establish for a particular stream, a stage-discharge relationship so that subsequent discharge determinations are merely obtained by measurement of the stage, with the discharge read from the rating curve
- The rating curve should be checked from time to time for accurate measurements

Stage-Discharge Relations

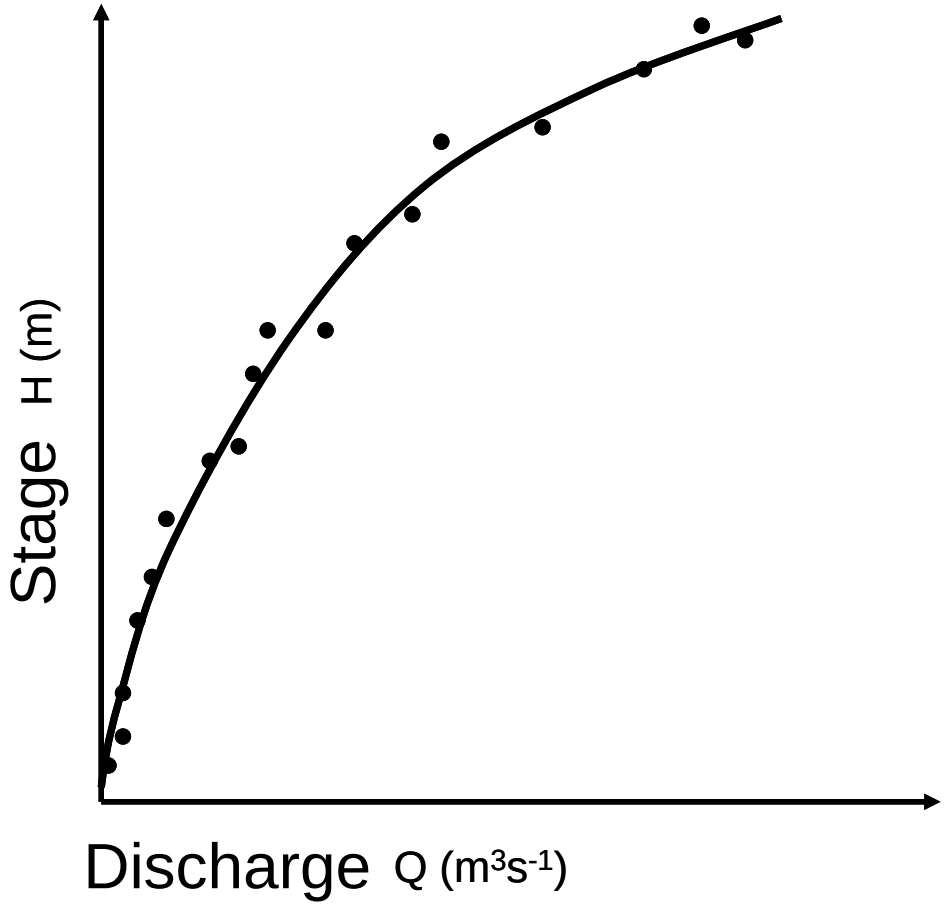


Fig. Sketch of rating curve